

NILAVA METYA

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DOB: December 30, 2001 (Age: 23)

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INTRODUCTION

Applied mathematician interested in borrowing tools from *convex geometry*, *random matrices*, *optimal transport*, *probability* to algorithmic problems in *machine learning*, especially those that require being robust to uncertain data.

EDUCATION

Rutgers, the State University of New Jersey - New Brunswick

Doctor of Philosophy in Mathematics | CGPA: 4.0/4.0

(passed written qualifying exams in first attempt just before program started)

Sep '22 – (expected) '28

Piscataway, New Jersey, USA

Chennai Mathematical Institute

Bachelor of Science (Honours) in Mathematics and Computer Science | CGPA: 9.72/10

Position: *Third* (out of 55 students)

(completed degree requirements in 2.5 years)

Aug '19 – May '22

Chennai, Tamil Nadu, India

Don Bosco School, Liluah

Indian School Certificate (ISC) 2019 | *Percentage*: 97.25%

Position: *First* in science stream (~ 55 students), *second* overall (~ 180 students)

Indian Certificate of Secondary Education (ICSE) 2017 | *Percentage*: 96.6%

Position: *First* in school (~ 180 students)

Apr '06 – Mar '19

Howrah, West Bengal, India

SELECTED GRADUATE-LEVEL COURSEWORK

- Quantum Computation
- Matrix Computations
- Statistics with R
- Algebraic Topology
- Complex Analysis
- Haskell

- Algorithm Design + Analysis
- Algebraic Geometry^{Princeton}
- Probability^{Rutgers}
- Data Mining^{Rutgers}
- Topological Data

- Analysis^{Rutgers}
- Convex Optimization^{Princeton}
- Convexities in Banach Spaces^{Princeton}
- Random Matrices^{Princeton}

PUBLICATIONS/PREPRINTS

2. N Metya, A Sinha. Temporal robustness in discrete time linear dynamical systems.
Submitted
1. G DePaul, S Hoşten, N Metya, I Nometa. Degrees of the Wasserstein distance to small toric models.
Journal of Algebraic Statistics

ATTENDANCE IN CONFERENCES/WORKSHOPS

Princeton Machine Learning Theory Summer School <i>Summer School</i> Princeton	Aug '24
Efficient Algorithms for High Dimensional Metrics <i>Workshop</i> DIMACS, Rutgers, New Brunswick	May '24
Bayesian Statistics and Statistical Learning <i>Workshop</i> IMSI, Chicago	Dec '23
Apprenticeship Week: Varieties from Statistics IMSI, Chicago	Oct '23
Algebraic Methods in Biochemical Reaction Networks MPI, Leipzig	Jun '23
Joint Mathematics Meetings Boston	Jan '23

ONGOING PROJECTS

Bilinear robustness in RL

Harmonic analysis and optimization over orbitopes

Rank Aggregation problem in the streaming setting

Regret bounds in contextual bandits with robustness considerations

ACADEMIC VISITS

School of Computing, National University of Singapore | *Prof Jonathan Scarlett*

May-Aug '25

SELECTED TALKS DELIVERED

Temporal robustness in discrete time linear dynamical systems. | 1 talk | INFORMS, Atlanta Oct '25

Temporal robustness in discrete time linear dynamical systems. | 1 talk | Indian Statistical Institute Aug '25

Temporal robustness in discrete time linear dynamical systems. | 1 talk | NUS, Singapore May '25

Wasserstein degrees of small toric models | 1 talk | CUNY Mina Rees Women and Math Conference Mar '25

Principal Components along Quiver representations | 1 talk | Rutgers course: *Computational Topology* Dec '23

Inference on growth process of a network | 1 talk | Rutgers course: *Data Mining* Dec '23

HONOURS AND AWARDS

Nominated by Rutgers Math department for SLMath summer school Jun '23

Summer school at Leipzig - awarded full travel funding. Only two students from Rutgers Math were fully funded by SLMath.

Academic Excellence Award at Rutgers Sep '22

Received a certificate and \$100 based on performance in Written Qualifying Exams.

Shriram Scholarship at CMI '19 – '22

Received institutional fee waiver and monthly stipend (based on entrance exam).

Ranked 4th nationally at the *Bachelor of Statistics (B.Stat.)* entrance examination '19

Indian Statistical Institute (ISI)

Informatics Olympiad '17, '18, '19

Selected among (approx) top 100-130 school students in India in Zonal Informatics Olympiad (ZIO).

Mathematical Olympiad Jan '18

Selected for Indian National Mathematical Olympiad (INMO) Training Camp | top 30 school students in West Bengal.

Program in Mathematics for Young Scientists (PROMYS) '18, '19, '20, '21

Awarded the Tara and Jasubhai Mehta Fellowship to PROMYS (among 5 Indian school students in 2018) based on a competitive process. Participated twice as a student ('18, '19) and twice as a counselor ('20, '21).

SERVICE

Coffee Hour (departmental socializing) | *Coordinator* | Rutgers Math Department Jan '25 -

Physics of Learning Theory (PhyLT) | *Organizer* | Rutgers Math Department Jan - Apr '25

Written Qualifying Exams: Algebra | *Instructor* | Rutgers Math Department Jun - Aug '23

Algebra 'N' Geometry Learning Seminar (ANGeLS) | *Organizer* | Rutgers Math Department Jan - Apr '23

Student Seminar | *Organizer* | Chennai Mathematical Institute Oct - Dec '22

ICO Camp (online) | *Combinatorics teacher* | CodeChef Nov '20

SKILLS

Languages Bengali (mother tongue), English (fluent), Hindi (fluent)

Programming JAVA, C++, Python, Haskell, R, HTML, SageMath, Macaulay2, MATLAB

Documentation $\LaTeX$, Microsoft Word